■ PS-A02

Voice-First Citizen Complaint System

Track: Web Dev + API Integration

Complete Execution Plan & Strategy Guide

■ Indian Language Support	■ Voice-First UX	■ Geo-Location	■ Admin Dashboard

This document provides a comprehensive project plan, technical architecture, sprint breakdown, tech stack recommendations, and risk mitigation strategies for building PS-A02 — a voice-driven civic complaint portal designed for India's semi-urban and rural citizens.

01 · Problem Understanding

Core Challenge

Millions of Indians in smaller cities cannot register civic complaints because existing portals are English-only, form-heavy, and not voice-friendly. The solution must allow a citizen to speak in any Indian language, auto-categorize the complaint, geo-tag it, and issue a tracking ID.

Key Constraints

Constraint	Impact	Our Approach
English-only portals	Excludes 80%+ rural users	Web Speech API with lang switching
Form-heavy UI	High dropout rate	Single-tap voice recording flow
No tracking	Zero accountability	UUID-based mock tracking IDs
Multiple departments	Misdirected complaints	Keyword ML categorization

02 · Solution Architecture

High-Level System Design

Frontend (HTML/CSS/JS) → Voice Input (SpeechRecognition API) → Transcript Text → Keyword Categorizer (JS + optional Claude AI) → Geo-location (Geolocation API) → Mock DB (localStorage / JSON) → Tracking ID Generator → Admin Dashboard

Component Breakdown

Layer	Component	Technology	Purpose
UI / Frontend	Citizen Portal	HTML5 + CSS3 + Vanilla JS	Voice recording, form display, confirmation
	Admin Dashboard	HTML5 + Chart.js	View all complaints, filter, update status
Voice Layer	Speech-to-Text	Web Speech API (SpeechRecognition)	Converts spoken complaint to text, multi-lang
AI/ML Layer	Categorizer	Keyword Map + Claude AI API (optional)	Maps complaint to department
Location Layer	Geo-tag	Browser Geolocation API	Captures lat/lng at complaint time
Data Layer	Mock Storage	localStorage / JSON file	Stores complaints for demo
ID Layer	Tracking ID	UUID v4 generator (JS)	Unique complaint reference number

03 · Recommended Tech Stack

Why This Stack?

Chosen for zero-setup deployment, browser-native APIs (no backend needed for MVP), and easy demo-ability within a hackathon timeframe. Everything runs client-side.

Category	Tool / Library	Why Chosen
Voice Input	Web Speech API (SpeechRecognition)	Built into Chrome; supports 'hi-IN', 'ta-IN', 'te-IN', 'bn-IN', etc.
Geo-location	Navigator Geolocation API	Native browser API, no library needed
Categorization	JS Keyword Map + Claude AI API	Fast keyword match as primary; Claude for edge cases
Frontend	HTML5 / CSS3 / Vanilla JS	No build step; easy to demo and share
Charts	Chart.js (CDN)	Lightweight, beautiful charts for admin view
Mapping	Leaflet.js + OpenStreetMap	Free map tiles; pin complaints on map
Storage (Mock)	localStorage	Persists across tabs in browser; zero setup
ID Generation	crypto.randomUUID()	Native JS method, no library needed
Hosting	GitHub Pages / Vercel	Deploy in minutes; shareable URL for demo

04 · Sprint Execution Plan (24–36 Hour Hackathon)

■ Phase 1 (0–4 hrs) *Foundation & Voice Setup*

- ✓ Setup project structure: index.html, style.css, app.js, admin.html
- ✓ Implement SpeechRecognition API with language selector (Hindi, Tamil, Telugu, Bengali, Marathi, English)
- ✓ Build the mic button UI — tap to record, real-time transcript display
- ✓ Test voice capture in at least 3 languages
- ✓ Add fallback: text input field if voice not available

■ Phase 2 (4–9 hrs) *Categorization + Geo + Storage*

- ✓ Build keyword categorization map for 6 departments: Water, Roads, Electricity, Sanitation, Health, Other
- ✓ Add Hindi + English keywords per department (transliterate common phrases)
- ✓ Integrate Navigator Geolocation API — capture lat/lng on complaint submit
- ✓ Implement localStorage schema: { id, transcript, category, lat, lng, timestamp, status }
- ✓ Generate UUID tracking ID using crypto.randomUUID()
- ✓ Build confirmation screen: show tracking ID + department assigned

■ Phase 3 (9–15 hrs) *Admin Dashboard + Map View*

- ✓ Build admin.html with complaint table (all entries from localStorage)
- ✓ Add status dropdown per complaint: Pending → In Progress → Resolved
- ✓ Integrate Leaflet.js map — pin each complaint with color-coded department markers
- ✓ Add Chart.js bar chart: complaints per department
- ✓ Add filter by department, date, and status
- ✓ Build simple stats panel: total complaints, resolved count, top department

■ Phase 4 (15–20 hrs) *Polish, AI Enhancement & Testing*

- ✓ Integrate Claude AI API for smart categorization of ambiguous complaints
- ✓ Add SMS-style confirmation animation after submission
- ✓ Make UI fully mobile-responsive (citizens will use phones)
- ✓ Test all 6 Indian languages — fix accent/dialect recognition issues
- ✓ Cross-browser test (Chrome required for Web Speech API)
- ✓ Seed 15–20 mock complaints in admin for demo impact

■ Phase 5 (20–24 hrs) *Presentation & Demo Prep*

- ✓ Deploy to GitHub Pages or Vercel — get a public URL
- ✓ Create PPT: Problem → Architecture Diagram → Live Demo → Impact Metrics
- ✓ Prepare 2-minute live demo script: record Hindi complaint → show categorization → open admin
- ✓ Add architecture slide with component flow diagram
- ✓ Rehearse demo 3 times — anticipate Wi-Fi issues (have offline fallback screenshots)

05 · Categorization Engine Design

Department Keyword Map

Primary approach: JavaScript keyword matching. Secondary: Claude AI API for edge-case classification. This dual approach ensures speed + accuracy within the hackathon demo.

Department	English Keywords	Hindi Keywords (Transliterated)	Icon
Water Supply	water, pipe, leak, supply, tap, drainage, sewage, flood	pani, nal, pipe, barbaadi, naali, baarish, sewage	■
Roads	road, pothole, street, footpath, bridge, repair, crack	sadak, khadda, rastha, pathar, pul, tooti	■
Electricity	electricity, power, light, wire, transformer, outage, voltage	bijli, light, current, wire, transformer, andhera	■
Sanitation	garbage, waste, trash, dustbin, cleanliness, gutter	kachra, safai, gandagi, naali, jhadu, dhalao	■
Health	hospital, doctor, medicine, ambulance, mosquito, disease	aspatal, dawai, doctor, machhar, bimari, swasthya	■
Other / General	(fallback when no keyword matches)	(Claude AI handles these)	■

Categorization Logic (Pseudocode)

```
function categorize(transcript) {
const text = transcript.toLowerCase();
for (const [dept, keywords] of Object.entries(KEYWORD_MAP)) {
if (keywords.some(kw => text.includes(kw))) return dept;
}
return callClaudeAPI(transcript); // fallback for ambiguous cases
}
```

06 · Data Flow & Storage Schema

Complaint Object Schema


```
{
"id": "550e8400-e29b-41d4-a716-446655440000", // UUID v4
"transcript": "Hamare mohalle mein pani nahi aa raha...",
"category": "Water Supply",
"language": "hi-IN",
"location": { "lat": 28.6139, "lng": 77.2090, "address": "New Delhi" },
"timestamp": "2026-05-24T10:30:00Z",
"status": "Pending", // Pending | In Progress | Resolved
"trackingId": "CMP-2026-3847"
}
```


User Flow (Citizen)

1	Open app → Select language (Hindi / Tamil / English...)
2	Tap ■ mic button → Speak complaint in chosen language
3	System shows real-time transcript → User reviews / edits
4	Auto-categorize complaint → Show predicted department
5	Allow geo-location → Capture coordinates
6	Submit → Generate tracking ID → Show confirmation screen

07 · Risk Register & Mitigations

Risk	Likelihood	Impact	Mitigation
SpeechRecognition API not working in demo browser	**HIGH**	**HIGH**	Pre-record audio clips as fallback; test Chrome only
Wi-Fi down during demo — geo/API calls fail	**MED**	**HIGH**	Store mock lat/lng; run Claude on device or skip AI layer
Accent/dialect issues in voice recognition	**HIGH**	**MED**	Test 3 accents; show text edit box so user can fix transcript
Mis-categorization of complaints	**MED**	**MED**	Add 'Wrong department?' link; user can reassign manually
localStorage limit exceeded with many mock entries	**LOW**	**LOW**	Clear and re-seed before demo; use JSON file if needed
Mobile SpeechRecognition support inconsistent	**MED**	**MED**	Demo on laptop Chrome; mobile as secondary show

08 · Suggested Team Roles (3–4 Members)

Role	Responsibilities	Hours Focus
Frontend Dev #1 (Voice + UI)	SpeechRecognition integration, citizen portal UI, language switcher, confirmation screen	Phase 1 & 2
Frontend Dev #2 (Admin + Maps)	Admin dashboard, Leaflet map, Chart.js analytics, status management	Phase 3
Full-Stack / AI	Categorization engine, Claude API integration, localStorage schema, UUID logic	Phase 2 & 4
PM / Designer + Presenter	UI/UX polish, mobile responsiveness, PPT creation, demo script, deployment	Phase 4 & 5

09 · Winning Edge — WOW Features

These extra features will differentiate your submission from other teams and make judges remember your demo.

- ■ Multi-language auto-detect — browser detects user's system language and pre-selects it
- ■ Reverse Geocoding — convert lat/lng to readable address using OpenStreetMap Nominatim (free API)
- ■ PWA (Progressive Web App) — add manifest.json so citizens can install it as an app
- ■ Status Notification Simulation — show toast when admin changes complaint status
- ■ Heat Map on Admin — use Leaflet.heat plugin to show complaint density areas
- ■ Text-to-Speech Confirmation — read back the tracking ID aloud so low-literacy users understand
- ■ Optional Photo Attach — let user take a photo of the issue to attach with complaint

10 · PPT Slide Outline (Deliverable)

Slide #	Title	Content
1	Problem Statement	Stats on civic complaint portal failure; language barrier visual
2	Our Solution	One-line pitch + system flow animation
3	Architecture Diagram	Component diagram: Voice → Categorizer → Geo → DB → Admin
4	Tech Stack	Icons grid of all technologies used
5	LIVE DEMO	Citizen view → Admin view → Map view (live or recorded)
6	Impact & Scalability	Who benefits, how to scale, languages supported
7	Team	Names, roles, GitHub/deployment link

■ **Key Success Metric: Judges should be able to speak a complaint in Hindi and see it appear correctly categorized in the admin dashboard within 10 seconds. That live moment wins the demo.**